

Bioinformatics

BioInf 2025/2026 1st Semester

André Lamúrias

a.lamurias@fct.unl.pt

Summary

- Course structure and assessment
- Introductions to Bioinformatics
 - Basic concepts
 - Historical Context
 - Definition
- Connection to Computer Science
- Core problems and applications

Overview

- Instructor: André Lamúrias
 - a.lamurias@fct.unl.pt
 - Office hours: Thursdays at 15h but confirm first by email
- Schedule:
 - Theoretical Lectures: Thursday 9-11
 - Practical Labs: Thursday 11-13
- CLIP: <https://clip.fct.unl.pt/>
- Moodle: <https://moodle.fct.unl.pt/course/view.php?id=9786>
 - Lecture and Lab pdfs
 - Project assignments
 - Additional documents and code

What is this course about?

- Main idea: understand methods to process biological data
- Basic concepts (biomolecules, biological processes, alignment algorithms, ...)
- Problems (multiple sequence alignment, motif discovery, genome recovery, ...)
- Applications (drug discovery, genome analysis, personalized medicine, ...)

Objectives

- Bioinformatics Fundamentals: Grasp core concepts, data types, and algorithms in bioinformatics
- Practical Implementation: Apply bioinformatics methods and access reference databases
- Method Evaluation: Critically assess the appropriateness of different bioinformatics techniques
- Autonomous Learning: Develop self-reliance in applying and advancing bioinformatics knowledge

Syllabus

- Introduction to Bioinformatics
 - Definition and overview of main application areas and challenges
 - Introduction to biomolecules
 - Data structures for biological sequences
 - Sequence processing
 - Algorithms for pattern finding
 - Pairwise sequence alignment
 - Fast database search
 - Multiple sequence alignment
 - Phylogenetic trees
 - Motif discovery
 - DNA sequencing
 - Genome sequencing technologies
 - Graph-based genome assembly
 - Reference-guided assembly
 - Biological systems and data integration
 - Biological databases and ontologies
 - Introduction to systems biology
 - Biological network analysis
 - Information extraction from literature
 - Recent developments in bioinformatics
-  1st test around here

Bibliography

- **Compeau**, Phillip. and Pavel A. Pevzner. Bioinformatics algorithms: an active learning approach. Active Learning Publishers, 2015
- **Rocha**, Miguel and Pedro G. Ferreira, Bioinformatics Algorithms: Design and Implementation in Python, Academic Press, 2018
- **Jones**, Neil C., and Pavel A. Pevzner. An introduction to bioinformatics algorithms. MIT press, 2004
- **Baxevanis**, Andreas D., Gary D. Bader, and David S. Wishart, eds. Bioinformatics (4th edition). John Wiley & Sons, 2020

Assessment

- Theoretical-Practical Assessment Component – 60% (minimum 9.5/20)
 - 2 tests or exam
- Project Assessment Component – 40% (minimum 9.5/20)
 - 2 group projects

Project Assignments

- Groups of 2 students
- Continuation of topics discussed in class
 - 1st project: Implement an algorithm based on a paper
 - 2nd project: Analyze a dataset of documents
- Each project assignment will be described in class
- Some of the labs will be dedicated to project support
- Submit report and source code
- Oral discussion about each project

Software

- Code examples will be in python
 - Recommended language because of biopython but you can use something else
- Rosalind: <https://rosalind.info/classes/enroll/e678b99afb/>
 - We will use this platform for some of the lab sessions
 - Create an account before next class

Lectures

- 2 lectures of 1 hour per week (break in the middle)
- Slides will be available on moodle (can be updated after class)
- Questions at any point during the lecture or afterwards by email or office hours
- Bibliography of each lecture given on the last slide
- Some slides will reference Lab exercises to be solved next

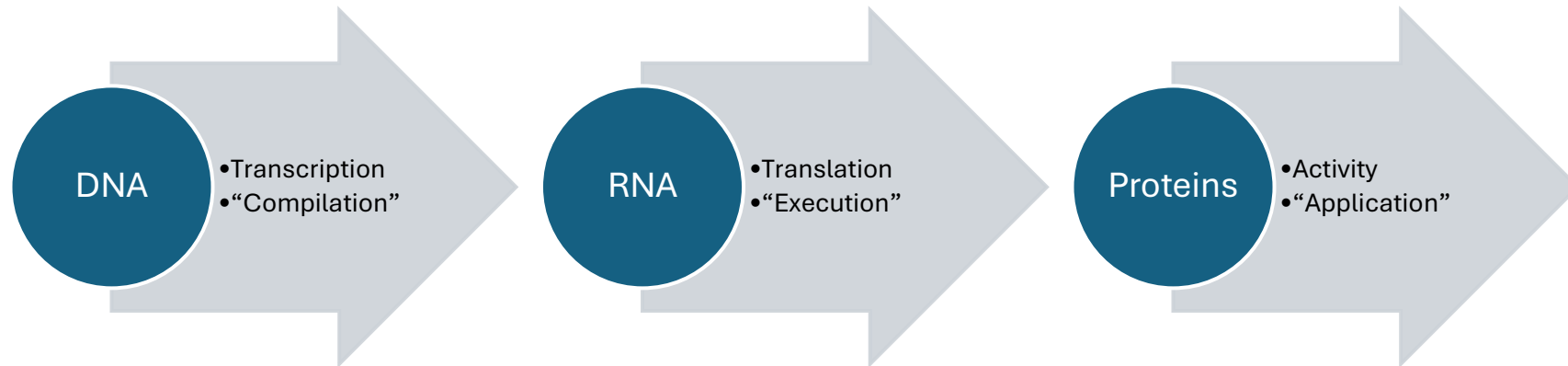
Labs

- Sets of problems to be solved during class
- Extra exercises for outside class
- Some exercises will be on Rosalind, others will be given in the slides
- For some Labs, some code will be provided to get you started
- These problems will help you understand the subjects of the theoretical lectures and develop the projects
- Using autocomplete tools (copilot) won't help you understand the subjects and might lead to wrong solutions

Bioinformatics

Bioinformatics

- Biological data is mostly sequential
- Genetic code -> programming language
- Central dogma of molecular biology:



Bioinformatics

Origins

- 1960s, using computers for molecular biology problems
 - Especially protein sequencing (Edman degradation)
 - Idea of molecules carrying biological information
- 1970s: Ben Hesper and Paulien Hogeweg
 - Bioinformatics: “the study of informatic processes in biotic systems”

Bioinformatics

Bioinformatics (original sense)

- Information flows in biological systems:
 - Information accumulated during evolution
 - Information from DNA to proteins
- As sub-discipline:
 - Biochemistry, Biophysics, Bioinformatics



Bioinformatics

Bioinformatics (broad sense)

- Application of computers to biological problems:
 - Data analysis and processing
 - Simulation
 - Knowledge representation and reasoning, AI
- Also computational biology:

“Computational biology and bioinformatics is an interdisciplinary field that develops and applies computational methods to analyse large collections of biological data, such as genetic sequences, cell populations or protein samples, to make new predictions or discover new biology. The computational methods used include analytical methods, mathematical modelling and simulation.”

<https://www.nature.com/subjects/computational-biology-and-bioinformatics>

Bioinformatics

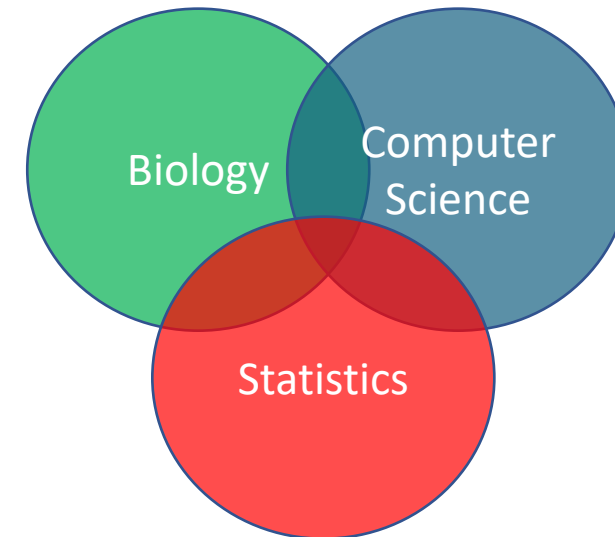
- Explosion of data
 - From sequencing projects: Human Genome Project
 - From other types of studies: gene expression, RNAseq, pharmacological
- ‘Omics: genomics, transcriptomics, proteomics, metabolomics, metagenomics, etc.
 - Large collections of molecules of the same type
 - Started with genomics: study of the genome
- Systems biology:
 - Study of complex biological systems as a whole instead of focusing on a single aspect



Bioinformatics

Bioinformatics is “interdisciplinary”

- Computer science:
 - Software engineering
 - Databases
 - Artificial Intelligence
 - Numerical methods
- Biology:
 - Genetics
 - Proteomics
 - Structural biochemistry
 - Evolution



Computer science

- Algorithms
- Data structures
- Machine Learning
- Databases and Big Data
- HPC and Cloud

Connection to other course subjects

- Database Systems
- Knowledge Representation and Reasoning
- Machine Learning and Deep Learning
- Unsupervised Learning
- Natural Language Processing and Search

Core problems

Core problems

- Sequence Alignment (Pairwise and multiple)

A T - G T T A T A
A T C G T - C - C

Input sequences

S₁: PEEMSVTSLDLTGGLPEATT ...
S₂: PEEMSVASLDLTGGLPEAST ...
S₃: SEELAAATALDLGAPSPAAA ...
S₄: PGPGLAEVRDLPGSTSAKE ...



	S ₁	S ₂	S ₃	S ₄
S ₁	-			
S ₂	76	-		
S ₃	45	53	-	
S ₄	31	27	43	-

Similarity matrix



S₁ PEEMSVTS-LDLTGGLPEATT ...
S₂ PEEMSVAS-LDLTGGLPEAST ...
S₃ SEELAAATALDLG-----APSP ...
S₄ PGPGLAEVRDLPG-----ST ...
... : .** .

Alignment

Core problems

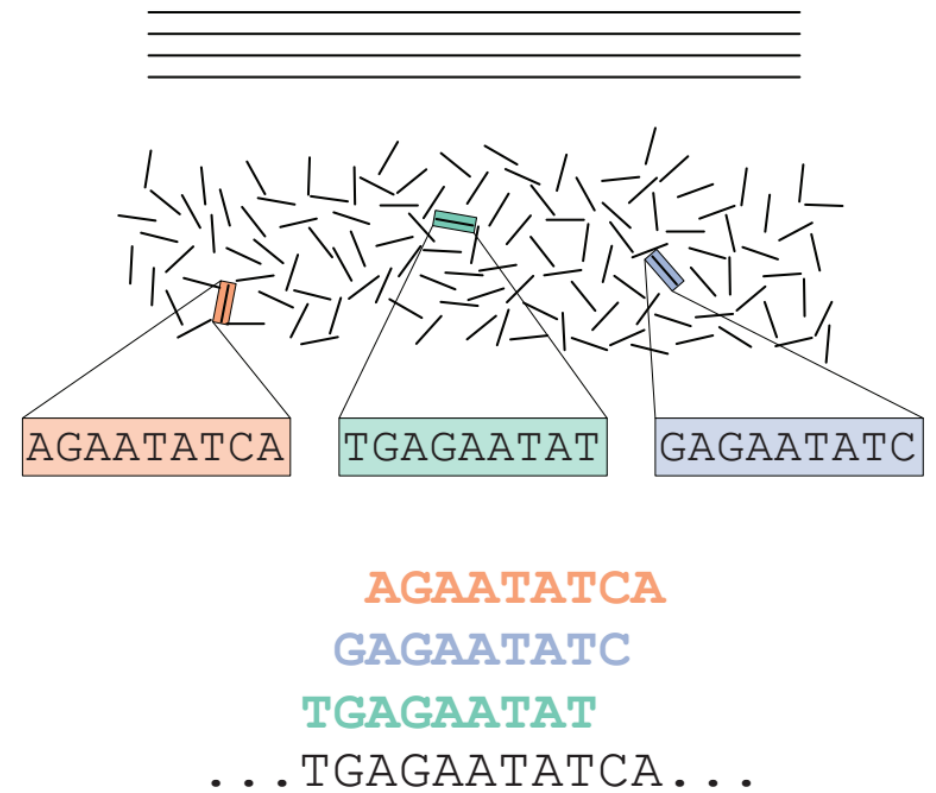
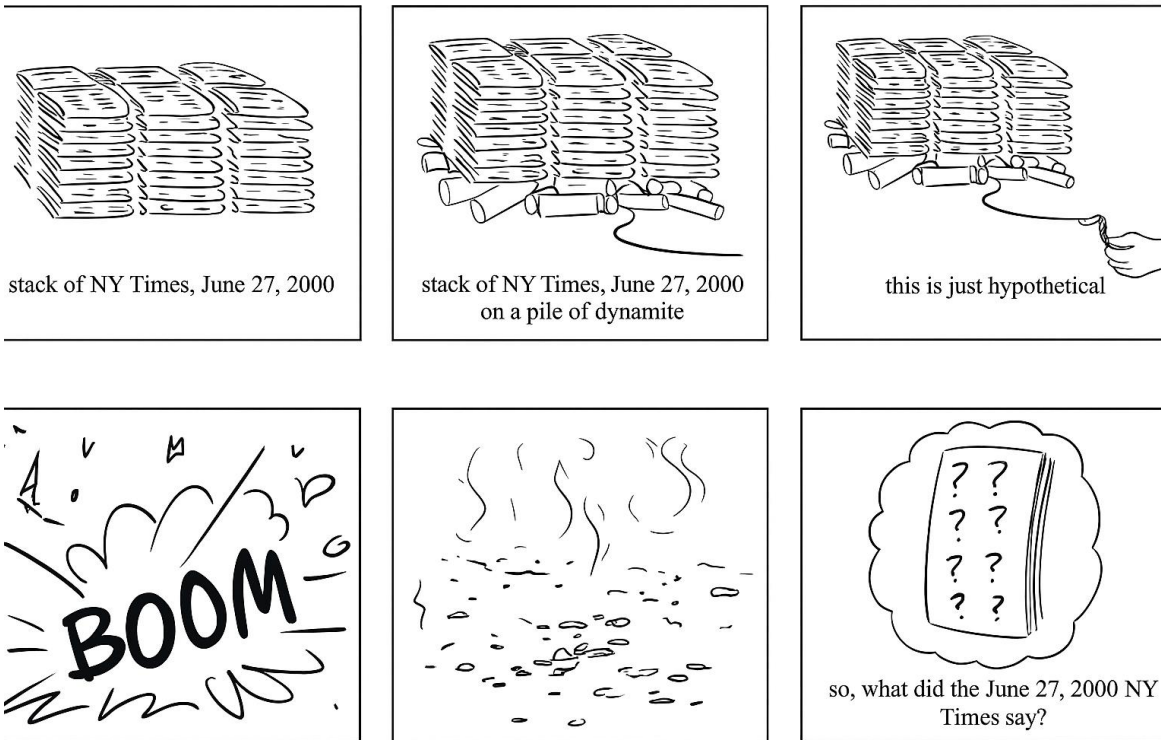
- Motif discovery and profiles

1	T	C	G	G	G	G	g	T	T	T	t	t
2	c	C	G	G	t	G	A	c	T	T	a	C
3	a	C	G	G	G	G	A	T	T	T	t	C
4	T	t	G	G	G	G	A	c	T	T	t	t
5	a	a	G	G	G	G	A	c	T	T	C	C
6	T	t	G	G	G	G	A	c	T	T	C	C
7	T	C	G	G	G	G	A	T	T	c	a	t
8	T	C	G	G	G	G	A	T	T	c	C	t
9	T	a	G	G	G	G	A	a	c	T	a	C
10	T	C	G	G	G	t	A	T	a	a	C	C



Core problems

- Genome Assembly



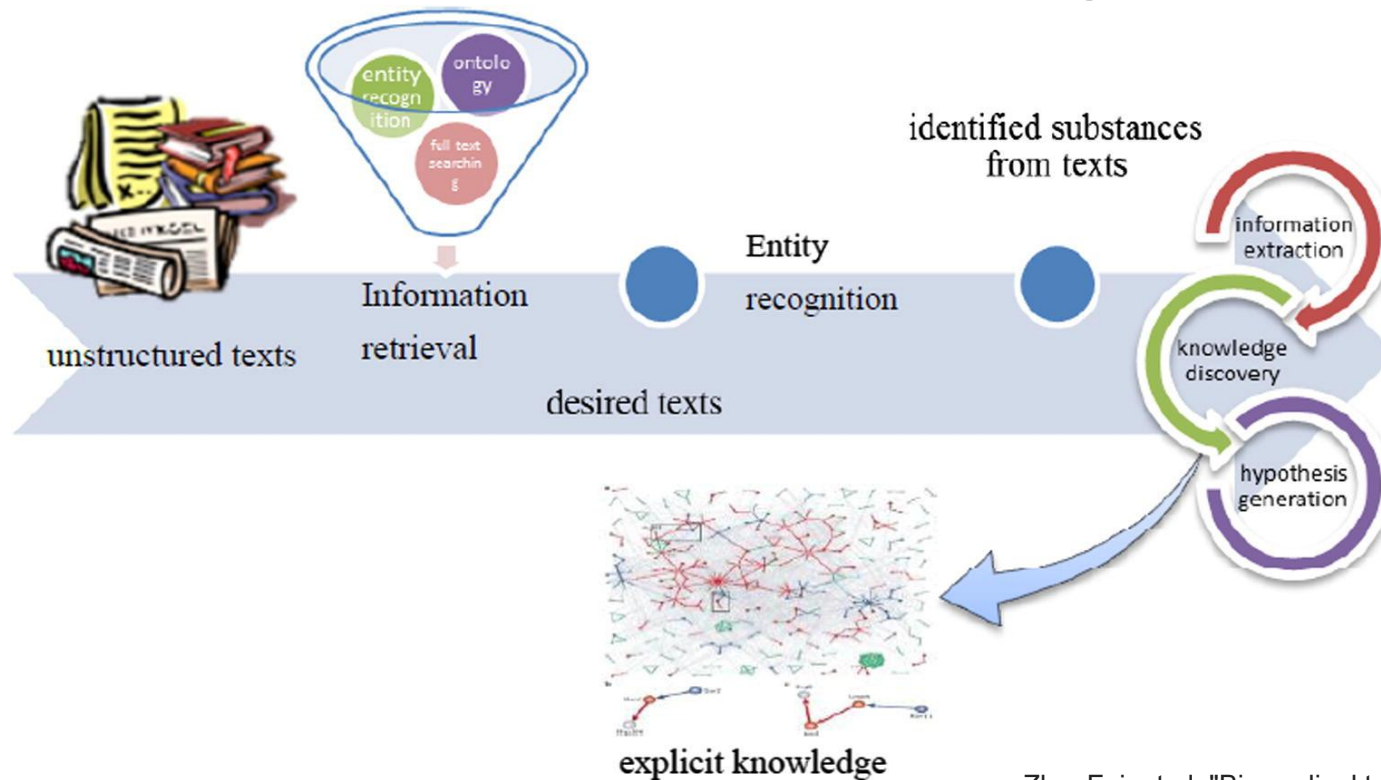
Core problems

- Database search

Sequences producing significant alignments									
Download ▼ Select columns ▼ Show 100 ▼ ?									
☒ select all 85 sequences selected GenBank Graphics Distance tree of results MSA Viewer									
	Description ▼	Scientific Name ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
☒	Methanosarcina barkeri str. Wiesmoor, complete genome	Methanosarcin...	1.529e+05	2.349e+05	100%	0.0	100.00%	4874815	CP009526.1
☒	Methanosarcina barkeri 227, complete genome	Methanosarcin...	13531	1.376e+05	91%	0.0	89.01%	4447980	CP009530.1
☒	Methanosarcina barkeri CM1, complete genome	Methanosarcin...	12859	1.098e+05	78%	0.0	89.73%	4501171	CP008746.1
☒	Methanosarcina barkeri MS, complete genome	Methanosarcin...	12859	1.335e+05	90%	0.0	89.73%	4533209	CP009528.1
☒	Methanosarcina vacuolata Z-761, complete genome	Methanosarcin...	12545	1.164e+05	79%	0.0	89.16%	4505752	CP009520.1
☒	Methanosarcina barkeri 3, complete genome	Methanosarcin...	9321	75331	72%	0.0	85.68%	4560446	CP009517.1
☒	Methanosarcina sp. Kolksee, complete genome	Methanosarcin...	8408	98592	74%	0.0	88.64%	4499114	CP009524.1
☒	Methanosarcina siciliae C2J, complete genome	Methanosarcin...	7592	66128	44%	0.0	89.75%	5427890	CP009508.1
☒	Methanosarcina flavescens strain E03.2 chromosome, complete genome	Methanosarcin...	4939	34848	41%	0.0	85.67%	3283688	CP032683.1
☒	Methanosarcina thermophila TM-1, complete genome	Methanosarcin...	4915	32547	40%	0.0	85.64%	3127379	CP009501.1
☒	Methanosarcina thermophila DNA, complete genome, strain: MT-1	Methanosarcin...	4915	32480	40%	0.0	85.64%	3194059	AP017646.1
☒	Methanosarcina thermophila CHTI-55, complete genome	Methanosarcin...	4915	32639	40%	0.0	85.64%	3130228	CP009502.1
☒	MAG: Methanosarcina sp. 795 genome	Methanosarcin...	4911	32685	40%	0.0	85.62%	3122414	CP011449.1
☒	Methanosarcina acetivorans str. C2A, complete genome	Methanosarcin...	4530	43460	29%	0.0	83.26%	5751492	AE010299.1
☒	MAG: uncultured Methanosarcinaceae archaeon isolate ENR-C_bin.1390 genome assembly, chrom...	uncultured Met...	4338	20902	30%	0.0	83.13%	3691339	QW818660.1

Core problems

- Information extraction and text mining

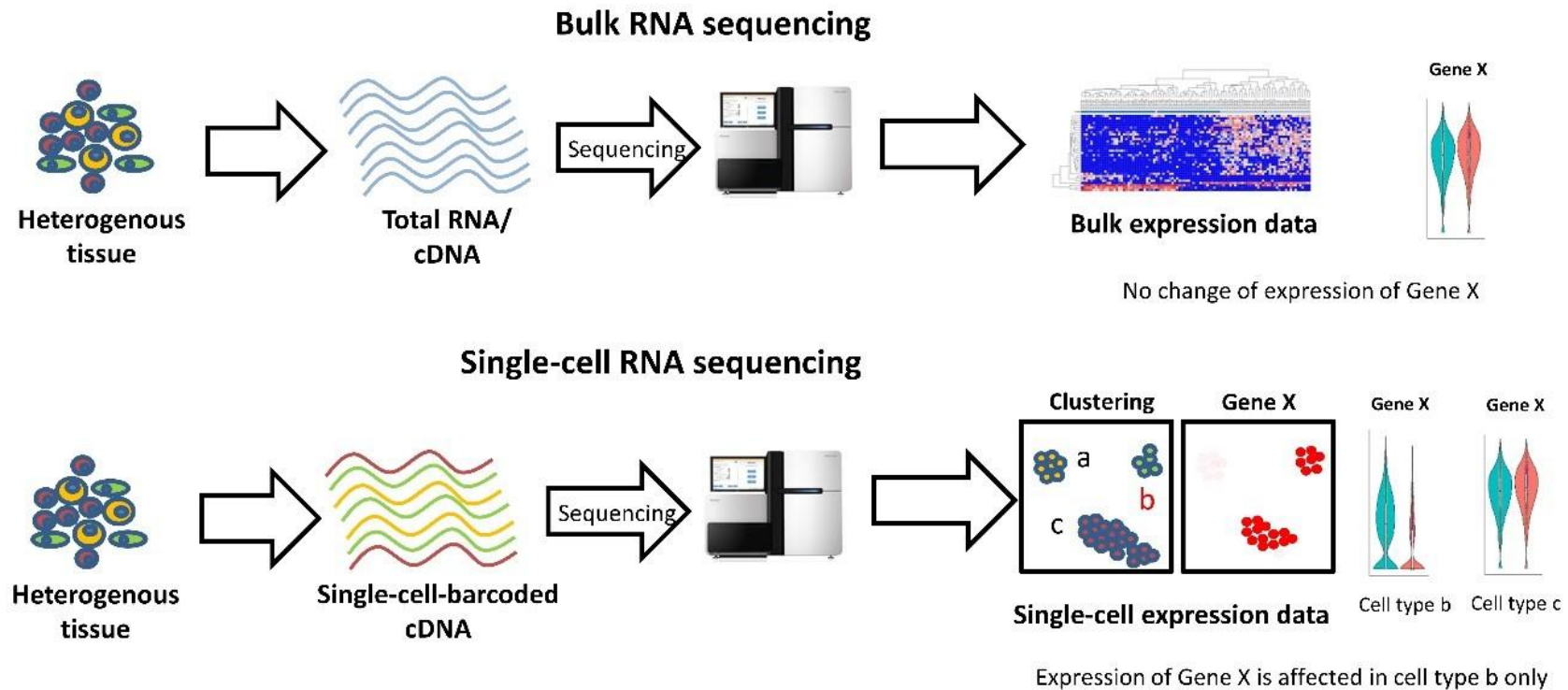


Zhu, Fei, et al. "Biomedical text mining and its applications in cancer research." *Journal of biomedical informatics* 46.2 (2013): 200-211.

Recent Developments

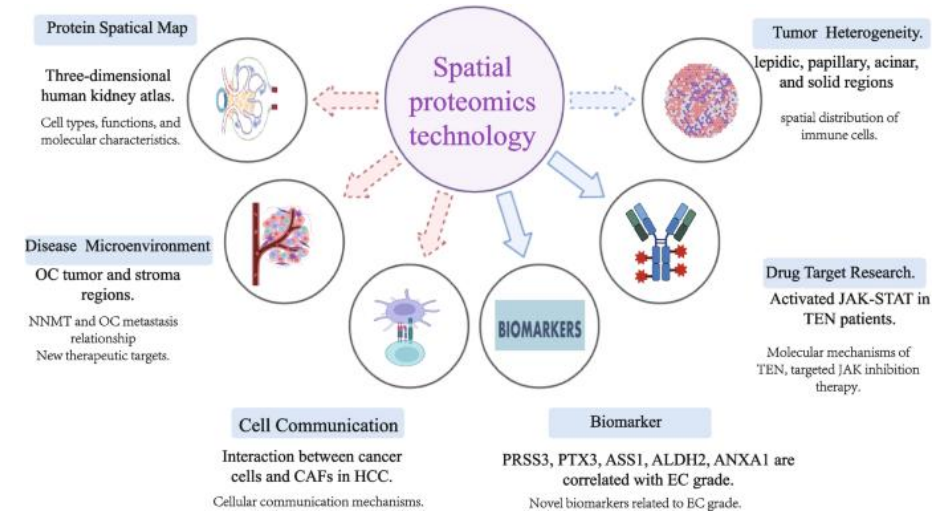
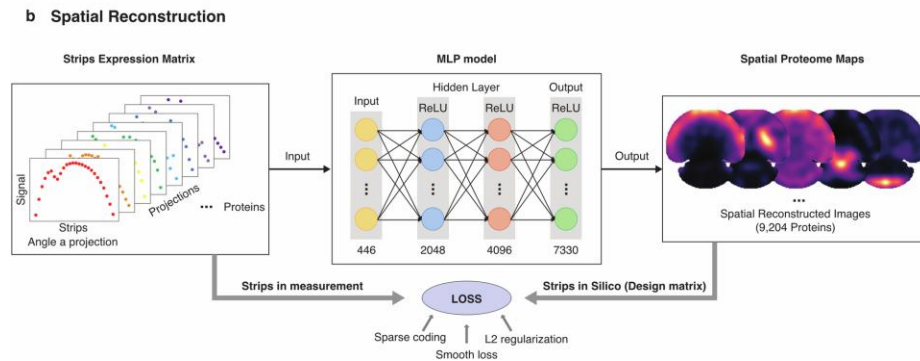
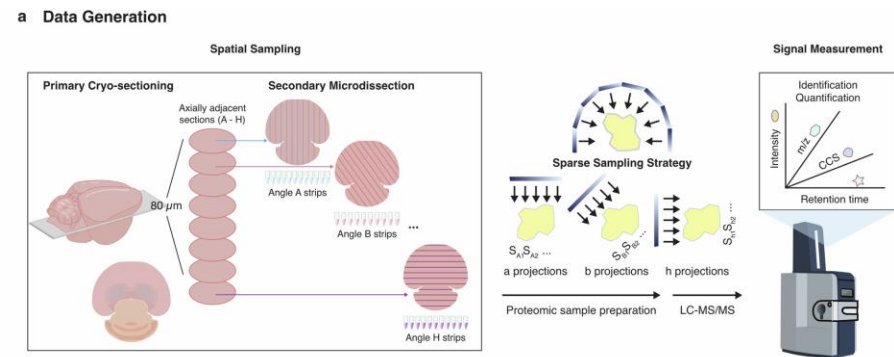
Recent Developments

- Single cell RNA



Recent Developments

- Spatial Proteomics



Nature methods of the year 2024:

<https://www.nature.com/articles/s41592-024-02565-3>

Challenges and opportunities

- **Challenges:**

- **Data Scale & Noise:** Biological data is messy and enormous
- **Complexity:** Biology is incredibly intricate
- **Interdisciplinary Gap:** The biggest challenge is bridging CS and biology

- **Opportunities:**

- **High Impact:** Solve critical problems in health, environment, food security
- **Cutting-Edge Research:** Many open problems where CS innovation is essential
- **New skills:** Learning to solve this type of problems will provide you with new transferable skills

Bibliography

- Rocha et al. 2018 Chapter 1